

AMAZON EC2 SHARED EFS

Step 1: Launch EC2-1

Go to:

AWS Console → EC2 → Launch Instance

Use:

- **Name:** ec2-efs-1
- **AMI:** Amazon Linux 2023
- **Instance type:** t3.micro
- **VPC:** Your VPC
- **Subnet:** us-east-1a
- **Auto-assign Public IP:** Enable
- **Key pair:** Your existing key pair

Security Group

Create/select:

Name: efs-sg

Inbound rules:

Type	Port	Source
------	------	--------

SSH	22	My IP
-----	----	-------

NFS	2049	efs-sg
-----	------	--------

Launch the instance.

Step 2: Launch EC2-2

Launch another EC2 with:

- **Name:** ec2-efs-2
- **AMI:** Amazon Linux 2023
- **Instance type:** t3.micro

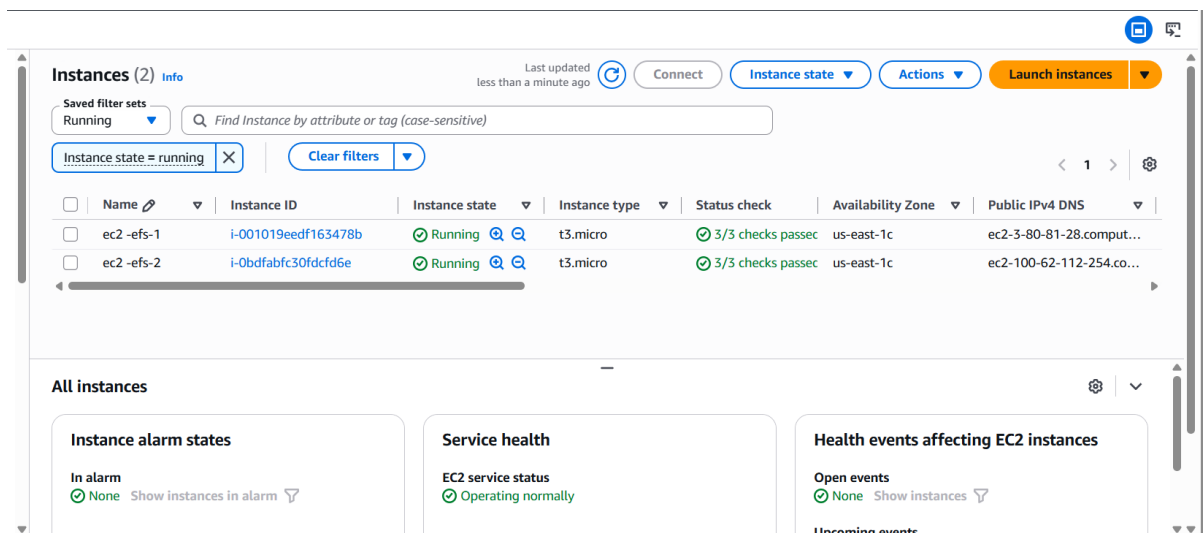
- **Same VPC**
- **Subnet:** us-east-1a
- **Public IP:** Enable
- **Same key pair**
- **Security Group:** efs-sg

So both instances should have:

VPC: Same

Availability Zone: us-east-1a

Security Group: efs-sg



Step 3: Create EFS

Go to:

AWS Console → EFS → File systems → Create file system

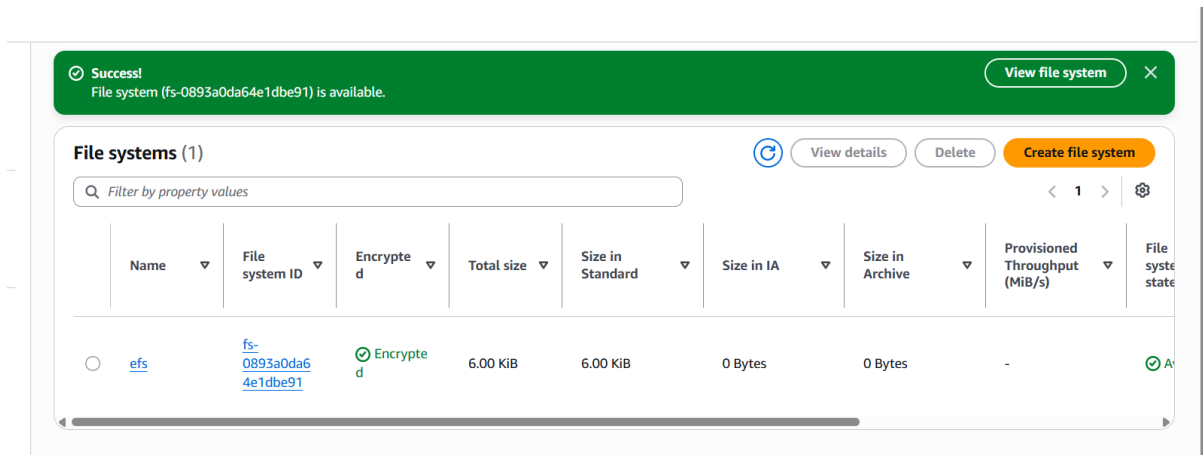
Choose **Customize**.

Set:

Name: efs-practice

VPC: Same VPC as EC2

Create the file system.



Step 4: Check EFS Mount Target

Open:

EFS → File systems → efs-practice → Network

You need a mount target like:

Availability Zone: us-east-1a

Subnet: EC2 subnet

Security Group: efs-sg

Status: Available

Important

Your EC2 is in:

us-east-1a

Therefore, make sure EFS has a mount target in:

us-east-1a

If it doesn't exist:

EFS → Network → Manage/Create mount target

Select:

VPC: Same VPC

Availability Zone: us-east-1a

Subnet: EC2 subnet

Security Group: efs-sg

Wait until:

Status: Available

Attach

Mount your Amazon EFS file system on a Linux instance. [Learn more](#)

☒ Mount via DNS

☐ Mount via IP

Using the EFS mount helper:

```
sudo mount -t efs -o tls fs-0f84a6ff7544fac70:/ efs
```

 client:

```
sudo mount -t nfs4 -o nfsvers=4.1,rsize=1048576,wsize=1048576,hard,timeo=600,retrans=2,noresvport fs-0f84a6ff7544fac70.efs.us-east-1.amazonaws.com:/ efs
```

See our user guide for more information. [Learn more](#)

[Close](#)

Step 5: Verify Security Group

Go to:

EC2 → Security Groups → efs-sg → Inbound rules

You must have:

NFS

TCP

2049

Source: efs-sg

This allows EC2 instances using efs-sg to communicate with EFS.

Step 6: Connect to EC2-1

From Windows PowerShell:

```
ssh -i new-key.pem ec2-user@<EC2-1-PUBLIC-IP>
```

You should get:

```
[ec2-user@ip-xxx ~]$
```



```
sudo mount -t nfs4 -o  
nfsvers=4.1,rsiz=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport fs-  
xxxxxxx.efs.us-east-1.amazonaws.com:/mnt/efs
```

Step 10: Mount EFS on EC2-1

Use the exact command from the EFS console.

Example:

```
sudo mount -t nfs4 -o  
nfsvers=4.1,rsiz=1048576,wsiz=1048576,hard,timeo=600,retrans=2,noresvport fs-  
0f84a6ff7544fac70.efs.us-east-1.amazonaws.com:/mnt/efs
```

Successful result

If successful, you will normally get **no output**.

You will simply return to:

```
[ec2-user@ip-xxx ~]$
```

Step 11: Verify the EFS mount

Run:

```
df -h
```

You should see an EFS filesystem mounted on:

```
/mnt/efs
```

You can also check:

```
mount | grep efs
```

Step 12: Create a file from EC2-1

Run:

```
cd /mnt/efs
```

Create a file:

```
echo "Hello from EC2-1" | sudo tee test.txt
```

Check it:

cat test.txt

Output:

Hello from EC2-1

Step 13: Connect to EC2-2

Open another PowerShell window:

ssh -i new-key.pem ec2-user@<EC2-2-PUBLIC-IP>

```
[ec2-user@ip-172-31-5-81 ~]$ ssh -i "ec2-key.pem" ec2-user@ec2-44-202-224-71.compute-1.amazonaws.com
Warning: Identity file ec2-key.pem not accessible: No such file or directory.
The authenticity of host 'ec2-44-202-224-71.compute-1.amazonaws.com (172.31.11.183)' can't be established.
ED25519 key fingerprint is SHA256:2ypKCykkJB57/Q5qEm08hMDPctLR4zC1kwCkRKqBi0s.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-44-202-224-71.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
ec2-user@ec2-44-202-224-71.compute-1.amazonaws.com: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-172-31-5-81 ~]$ sudo dnf install -y nfs-utils
Amazon Linux 2023 Kernel Livepatch repository                                     566 kB/s | 69 kB    00:00
Package nfs-utils-1:2.5.4-2.rc3.amzn2023.0.3.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-5-81 ~]$ sudo mkdir -p /mnt/efs
[ec2-user@ip-172-31-5-81 ~]$ sudo mount -t nfs4 -o nfsvers=4.1,rsz=1048576,wsz=1048576,hard,timeo=600,retrans=2,noresvport fs-0f84a6ff7544fac70.efs.us-east-1.amazonaws.com:/ /mnt/efs
[ec2-user@ip-172-31-5-81 ~]$ df -h
Filesystem                                Size  Used Avail Use% Mounted on
devtmpfs                                 420M   0   420M   0% /dev
tmpfs                                    457M   0   457M   0% /dev/shm
tmpfs                                    183M 444K  183M   1% /run
efivarfs                                128K 2.7K  121K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p1                          8.0G  1.7G   6.3G  22% /
tmpfs                                    457M   0   457M   0% /tmp
/dev/nvme0n1p128                         10M  1.3M   8.7M  13% /boot/efi
fs-0f84a6ff7544fac70.efs.us-east-1.amazonaws.com:/ 8.0E   0   8.0E   0% /mnt/efs
tmpfs                                    92M   0    92M   0% /run/user/1000
[ec2-user@ip-172-31-5-81 ~]$ cat /mnt/efs/test.txt
hello from EC2-1
[ec2-user@ip-172-31-5-81 ~]$ |
```

Step 14: Install NFS utilities on EC2-2

sudo dnf install -y nfs-utils

Step 15: Create mount directory on EC2-2

sudo mkdir -p /mnt/efs

Step 16: Mount the SAME EFS on EC2-2

Use the same EFS DNS command:

sudo mount -t nfs4 -o

nfsvers=4.1,rsz=1048576,wsz=1048576,hard,timeo=600,retrans=2,noresvport fs-0f84a6ff7544fac70.efs.us-east-1.amazonaws.com:/ /mnt/efs

Again, successful mounting normally gives **no output**.

Step 17: Verify from EC2-2

Run:

```
df -h
```

Then:

```
ls -la /mnt/efs
```

You should see:

```
test.txt
```

Now:

```
cat /mnt/efs/test.txt
```

Expected:

```
Hello from EC2-1
```

EFS shared storage is working.
